



Answer: Yes. In fact the constraint is $x_1 + a_2 x_2 = 1$ with objective $T = x_1 + x_2$. Since from the code one finds $\sum_{k=3}^{\infty} (1/2)^k = (1/2)^3 / (1 - 1/2) = 1/4$ ¹, it follows $a_2 = 2 \cdot (1/4) + 1/2 = 1$ (plus a negligible amount), so $a_2 > 1$. Thus the faster (second) therapy is optimal and the minimiser is $(x_1, x_2) = (0, 1/a_2)$. The ∞ -norm distance from $(0.1, 0.9)$ to $(0, 1/a_2)$ is $\max(|0.1 - 0|, |0.9 - 1/a_2|)$. Since $1/a_2 \approx 1$, this is $\max(0.1, |0.9 - 1|) \approx 0.1$, which is < 0.15 . Hence $(0.1, 0.9)$ is within 0.15 of the true minimiser.

Sources: Computation of the infinite geometric series ¹ confirms the series sum used to find a_2 .

¹ Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series